

LAB#6: POWER DEFECT MEASUREMENT LABORATORY

LABORATORY OBJECTIVES:

1. Manipulate control rods to change net reactivity (ρ_{net}) in the PULSTAR reactor core, and establish criticality at preset power levels.
2. Observe changes to reactivity parameters as reactor power increases.
3. Plot **Power Defect (pcm)** -vs- **Power (kW)** at each power level using the Reg. Rod Integral Worth Curve.
4. Measure the **Power Coefficient α_{power} (pcm/MW)** and determine the **Point of Adding Heat (kW)**.

EQUIPMENT:

- PULSTAR Reactor.
- Current Regulating (Reg.) Integral Rod Worth Curve.

PRECONDITIONS:

1. Reactor Daily Startup Checklist completed and signed by Designated Senior Reactor Operator (DSRO).
2. Reactor startup has been authorized by DSRO.
3. Licensed Reactor Operator is on duty at the reactor control console.

EQUATIONS:

$$\Delta\rho_{\text{rods}} = \rho_{\text{rods}} - \rho_{\text{rods-initial}}$$

Where:

 $\Delta\rho_{\text{rods}}$ = Power Defect (pcm) ρ_{rods} is the rod reactivity value (pcm) obtained at each power level. $\rho_{\text{rods-initial}}$ is the initial rod reactivity value (pcm) obtained at 1 kW.

$$\alpha_{\text{power}} = \alpha_{T\text{mod}} + \alpha_{T\text{fuel}}$$

Where:

 $\alpha_{T\text{mod}}$ is -3.9 pcm/°F (or -27 pcm/MW) for PULSTAR $\alpha_{T\text{fuel}}$ is -1.6 pcm/°F (or -303 pcm/MW) for PULSTAR**PROCEDURE:**

1. Bring the reactor critical and establish critical gang rod positions at a reactor power of 1 kilowatt (kW).
2. Withdraw control rods and place the reactor on a positive period of not less than 15 seconds.
3. Allow power to increase until the desired power level is reached (see Table #1 on next page) and then insert rods to reestablish criticality.
4. Establish criticality at each of the power levels given on Table #1 and record the Reg. Rod Position (inches) at each power level.
5. Repeat steps #2 through #4 until a reactor power of 950kW is reached.

Table #1: Reactivity –vs- Reactor Power

Linear Channel Power (kW)	Reg. Rod Position (inches)	Reg. Rod Reactivity Worth ρ_{rods} (pcm)	Power Defect (pcm) <i>$\Delta\rho_{\text{rods}}$ relative to value @ 1kW</i>
1.0			
5.0			
10			
25			
50			
75			
100			
250			
500			
750			
950			

ANALYSIS:

1. Using the Reg. Integral Rod Worth Curve, determine and record ρ_{rods} (pcm) for each power level.
2. For each power level, calculate **Power Defect (pcm)** by calculating the value for $\Delta\rho_{\text{rods}}$ using the initial value for ρ_{rods} at 1kW.
3. Plot **Power Defect (pcm) – vs- Power (kW)**
4. Determine the Power Defect at 1000 kW and compare with the established value for the PULSTAR **Power Coefficient** $\alpha_{\text{power}} = -330\text{pcm/MW}$.
5. Determine **Point of Adding Heat (kW)**.